Biology — Class 12 (PCTB, English Medium)

Solved Exercises | Punjab Curriculum and Textbook Board | Edition 2020

NOTE: Chapters 15-26 (the complete supplied 2020-edition file) are covered in full except: diagrams described in words (Ch. 15 nephron; Ch. 16 skull and appendicular skeleton; Ch. 17 nerve impulse), and one exercise number not shown on the exercise pages of most chapters (flagged inside each chapter). No Chapter 27 exists in the supplied file; a newer-syllabus book was NOT mixed in.

Chapter 15: Homeostasis

Exercise Q2 — Fill in the blanks

Osmoregulation is the ability of an organism to regulate its fluid contents.

The detoxification of ammonia to urea requires the precursor ornithine.

In the kidney, the nephron is closely associated with a network of capillaries .

In insects, salt and water reabsorption takes place in the rectum .

The antidiuretic hormone acts on the collecting tubules to promote reabsorption of water in the vertebrate nephron.

The nephrons arranged along the border of cortex and medulla, with tubular systems looping deep in the inner medulla, are called juxtamedullary nephrons.

The non-surgical procedure of removing kidney stones is termed as lithotripsy .

Hypothalamus is the homeostatic thermostat in humans.

Exercise Q3 — Short questions

Differentiate between osmoconformers and osmoregulators.

Osmoconformers allow their internal salt concentration to change with the surrounding water.

Most marine invertebrates are osmoconformers and save energy this way.

Osmoregulators keep their internal concentration constant by active transport, whatever the surroundings.

Freshwater animals, marine bony fish and land vertebrates are osmoregulators.

Define anhydrobiosis with an example.

Anhydrobiosis is survival in an almost completely dry, dormant state.

The organism loses nearly all its body water and its metabolism almost stops.

Tardigrades (water bears) and the larvae of some midges survive years in this state.

When water returns, they rehydrate and resume normal life.

Why does filtration take place only at the glomerulus part of the nephron and nowhere else?

The glomerulus is a tuft of capillaries with very high blood pressure.

Its capillary walls are fenestrated and the capsule has filtration slits.

Together they form a filter that lets water and small solutes pass but holds back cells and proteins.

No other part of the nephron has this pressure and this porous structure, so filtration occurs only there.

What is renal failure?

Renal failure is the condition in which the kidneys can no longer filter blood properly.

Waste products like urea and creatinine accumulate in the blood.

Water and salt balance is disturbed, and blood pressure rises.

Severe renal failure is treated by dialysis or kidney transplant.

Give one main adaptation in plants to high temperature and one to low temperature.

Against high temperature, many desert plants have thick waxy cuticles and small

leaves to reduce water loss.

Some also roll their leaves or close stomata during the hottest hours.

Against low temperature, alpine plants produce antifreeze proteins in their cells.

Their compact, low growth form also protects them from freezing winds.

Exercise Q3 — Short questions (continued)

Q3(iv). Mention two metabolic altered states that generally (70%) cause kidney stone formation.

About 70% of kidney stones are calcium oxalate stones, formed when urine becomes supersaturated.

The two main metabolic states behind them are hypercalciuria (too much calcium in the urine) and hyperoxaluria (too much oxalate in the urine).

High urinary calcium and oxalate crystallise together in the kidney tubules and grow into stones.

Exercise Q3(iv), Q4 — Extensive questions

Q4(i). Discuss the nature of excretory products in animals in various habitats, specifically in association with water availability.

Animals excrete nitrogenous waste in the form that fits their water supply.

Aquatic animals, with plenty of water, excrete highly toxic ammonia directly (ammonotelic), for example most fishes.

Land mammals convert ammonia to less toxic urea, which needs only moderate water (ureotelic), for example humans.

Birds, reptiles and insects, which must save water, excrete almost solid uric acid (uricotelic), which needs very little water.

Q4(ii). Account the excretory system in earthworm.

The earthworm excretes through segmental metanephridia, one pair in most

segments.

Each metanephridium has a ciliated funnel (nephrostome) opening into the coelom, a coiled tubule surrounded by blood capillaries, and a nephridiopore opening to the outside.

Waste from the blood and coelomic fluid is collected, useful substances are reabsorbed in the tubule, and the remaining fluid leaves through the nephridiopore.

Chloragogen cells around the gut also store and excrete waste materials.

Q4(iii). Highlight the role of liver as an excretory organ.

The liver is an excretory organ because it removes nitrogenous waste and other harmful substances from the blood.

It deaminates excess amino acids, producing ammonia, which it converts to urea through the ornithine cycle for excretion by the kidneys.

It breaks down old red blood cells, forming bile pigments (bilirubin), which are excreted in bile into the intestine.

It also detoxifies drugs, alcohol and poisons, and excretes cholesterol and excess hormones in bile.

Q4(iv). Draw a labelled diagram of a vertebrate nephron with all blood supply. State the function of each part.

DIAGRAM DESCRIBED IN WORDS (the builder has no nephron diagram type): a vertebrate nephron consists of the glomerulus (a tuft of capillaries), Bowman's capsule enclosing it, the proximal convoluted tubule, the loop of Henle (descending and ascending limbs), the distal convoluted tubule, and the collecting duct.

Blood supply: the afferent arteriole brings blood to the glomerulus, the efferent arteriole leaves it, and peritubular capillaries (vasa recta around the loop) surround the tubule.

Functions: the glomerulus filters blood under pressure; Bowman's capsule collects the filtrate; the proximal tubule reabsorbs glucose, salts and most water; the loop of

Henle concentrates urine; the distal tubule and collecting duct adjust salt and water under ADH control.

Q4(v). Describe thermoregulatory strategies in mammals including humans in cold temperature.

In cold temperature, mammals reduce heat loss by vasoconstriction of skin blood vessels, so less warm blood reaches the surface.

They raise fur (or rely on a fat layer under the skin, and blubber in whales and seals) to trap insulating air and reduce heat loss.

They increase heat production by shivering thermogenesis (rapid muscle contractions) and non-shivering thermogenesis (thyroid hormones and brown fat).

Behaviour such as curling up, huddling and seeking shelter also conserves heat.

Q4(vi). Discuss excretion in plants.

Plants have no special excretory organs, but they remove waste in several ways.

Gaseous waste (carbon dioxide and oxygen) leaves through stomata during respiration and photosynthesis, and through lenticels in bark.

Excess water and dissolved salts are lost by transpiration and guttation, and waste products are stored in old leaves that fall, in bark, and in vacuoles.

Some plants excrete resins, gums, latex and tannins, which also protect them from herbivores.

Q4(vii). Discuss some kidney problems with their cures.

Kidney stones: hard deposits that block urine flow and cause pain; cured by lithotripsy (shock waves that break the stone), medicines, or surgery.

Renal (kidney) failure: the kidneys can no longer filter blood; treated by dialysis (artificial filtering) or kidney transplant.

Urinary tract infections: bacterial infections causing burning urination; cured by antibiotics and plenty of fluids.

Nephritis (inflammation of the kidney) and kidney damage from diabetes or high

blood pressure are treated by controlling the underlying disease and, in severe cases, dialysis.

Chapter 15 Q1 — flagged, not shown on the exercise pages.

FLAGGED — BOOK NUMBERING NOTE: the Chapter 15 exercise pages as extracted begin at Q2; no Q1 appears on those pages.

Q1 is therefore not guessed here; this is a render-verified book numbering skip in the book.

Chapter 16: Support and Movement

Exercise Q1 — Fill in the blanks

Each muscle is enclosed by a membrane known as epimysium .

Osteoporosis is caused by the decrease in the level of oestrogen .

The moulting is controlled by a hormone ecdysone .

Glycogen is stored in the muscle cell as reserve food.

Collenchymatous cells lack lignin in their primary wall.

There are seven vertebrae in the neck region of mammals.

The most abundant proteins in the muscle are actin and myosin .

Tendons connect a muscle to a bone.

Thick muscle filament is composed of myosin .

Exercise Q2 — True or False

The shoulder joint is a hinge joint.

False

Tendons connect bones together at joints.

False

Arthritis often accompanies aging.

True

Calcium provides energy for muscle contraction.

False

Most of the sclerenchymatous cells are non-living.

True

Visceral muscles are striated, involuntary and smooth.

False

Exercise — Short questions (selection)

Explain the sliding filament theory of muscle contraction.

A muscle fibre contracts when its thin actin filaments slide over the thick myosin filaments.

Myosin heads attach to actin, pull it inward, detach and reattach, using energy from ATP.

The sarcomere shortens, but the filaments themselves do not change length.

Calcium ions released from the sarcoplasmic reticulum expose the binding sites on actin to start the cycle.

What is osteoporosis? Why does it occur?

Osteoporosis is a disease in which bones lose minerals and become weak and porous.

It occurs when bone breakdown is faster than bone formation.

Lack of calcium, vitamin D and oestrogen after menopause speeds up this loss.

The bones then fracture easily, especially at the hip, wrist and spine.

Give the functions of the human skeleton.

The skeleton gives the body its shape and supports its weight.

It protects soft organs: the skull protects the brain and the ribs protect the heart

and lungs.

It provides surfaces for muscle attachment, so movement is possible.

Bones also store calcium and produce blood cells in their marrow.

Exercise Q4 — Short questions

Q4(i). What is the cause of cramps?

Cramps are sudden, involuntary and painful contractions of a muscle that do not relax.

They are caused by overuse, prolonged contraction, loss of salts (especially calcium and potassium) in sweat, and accumulation of lactic acid.

They usually pass when the muscle is stretched and rested and salts and water are replaced.

Q4(ii). What is the difference between tetanus and muscle tetany?

Tetanus, in muscle physiology, is a sustained, fused contraction produced when a muscle is stimulated so rapidly that it cannot relax between stimuli.

Muscle tetany is a pathological state of involuntary muscle spasms and twitches, usually caused by low blood calcium (hypocalcaemia), which makes nerves and muscles over-excitable.

In short, tetanus is a normal sustained contraction response, while tetany is a disease-like condition of uncontrolled spasms.

Q4(iii). What is a ligament?

A ligament is a tough band of fibrous connective tissue that joins one bone to another bone at a joint.

Ligaments hold the joint together, limit its movement to the correct direction, and prevent dislocation.

They are slightly elastic, so they allow a little movement while keeping the joint stable.

Q4(iv). What is nutation?

Nutation is the spontaneous circular or elliptical movement of a growing plant organ, such as a shoot tip or tendril.

The growing tip bends in turn to all sides, sweeping a circle as it grows, which helps tendrils find a support to climb.

It is a growth movement caused by unequal growth on different sides of the organ.

Q4(v). How many ribs do not attach with the sternum?

Two pairs of ribs, the 11th and 12th pairs, do not attach to the sternum at all; they are called floating ribs.

Ribs 8, 9 and 10 do not attach directly to the sternum either; their cartilages join the cartilage of the rib above.

Only the first seven pairs (true ribs) attach directly to the sternum by their own costal cartilages.

Q4(vi). How is rickets produced?

Rickets is a disease of growing children in which bones become soft, weak and deformed.

It is produced by deficiency of vitamin D, which is needed for the absorption of calcium and phosphorus from the intestine.

Without enough calcium, new bone cannot harden properly, so the legs bow and the chest and skull deform.

Q4(vii). What is the cause of tetanus?

Tetanus (the disease) is caused by the bacterium Clostridium tetani, which enters through wounds.

The bacterium produces a toxin that blocks the inhibitory signals to muscles, so the muscles contract continuously and cannot relax.

This causes lockjaw, rigid spasms and, without treatment, death; it is prevented by tetanus vaccination.

Q4(viii). How is muscle fatigue produced?

Muscle fatigue is the temporary loss of a muscle's ability to contract with full force after prolonged activity.

It is produced by depletion of ATP and glycogen, and by accumulation of lactic acid and other waste products in the muscle.

Lactic acid lowers the pH inside the muscle fibres and interferes with contraction, until rest and oxygen restore the muscle.

Q4(ix). Distinguish between: (a) axial and appendicular skeleton, (b) phototactic and chemotactic stimulus, (c) osteocytes and osteoblast, (d) brachialis and brachioradialis, (e) origin and insertion of muscles, (f) bone and cartilage, (g) troponin and tropomyosin.

(a) Axial skeleton is the central skeleton (skull, vertebral column, ribs, sternum); appendicular skeleton is the limbs and girdles that attach to it.

(b) Phototactic stimulus is movement in response to light; chemotactic stimulus is movement in response to a chemical.

(c) Osteocytes are mature bone cells that maintain bone tissue; osteoblasts are bone-forming cells that build new bone.

(d) Brachialis lies under the biceps and flexes the elbow; brachioradialis lies in the forearm and also helps flex the elbow, especially in mid-position.

(e) Origin is the fixed attachment of a muscle (usually to the less movable bone); insertion is the movable attachment, which moves when the muscle contracts.

(f) Bone is hard, mineralised connective tissue with calcium salts; cartilage is softer, flexible tissue without blood vessels, which cushions joints.

(g) Troponin is the protein complex that binds calcium and controls contraction; tropomyosin is the filament protein that blocks the myosin-binding sites on actin until calcium arrives.

Exercise Q5 — Extensive questions

Q5(i). What are the disadvantages of exoskeleton?

An exoskeleton limits growth, so the animal must moult, and during moulting it is soft and vulnerable to predators.

It is heavy relative to the animal's size, which limits how large the animal can grow, and it restricts the size of muscles inside it.

It also wears out and must be replaced, which costs energy and materials.

Q5(ii). What is the sliding filament model? What does it explain?

The sliding filament model says that muscle contraction happens when thin actin filaments slide over thick myosin filaments.

The filaments themselves do not shorten; instead, myosin heads attach to actin, pull it, detach and reattach, so the sarcomere shortens.

The model explains why the I-band and H-zone shorten during contraction while the A-band stays the same length.

Q5(iii). Describe a hinge joint and explain how it is moved by antagonistic muscle.

A hinge joint allows movement in one plane only, like a door hinge; examples are the elbow and knee joints.

It is moved by antagonistic muscle pairs: when one muscle contracts, its partner relaxes, and vice versa.

At the elbow, the biceps contracts to flex the arm while the triceps relaxes, and the triceps contracts to extend the arm while the biceps relaxes.

Q5(iv). Define joints. How are they classified? Explain.

A joint is a place where two or more bones meet and allow (or restrict) movement.

Joints are classified as fibrous (immovable, for example skull sutures), cartilaginous (slightly movable, for example between vertebrae), and synovial (freely movable, for example shoulder, elbow and knee).

Synovial joints have a joint cavity with synovial fluid, cartilage covering the bone

ends, and ligaments holding the bones together.

Q5(v). Explain appendicular skeleton with the help of a diagram (described in words).

DIAGRAM DESCRIBED IN WORDS: the appendicular skeleton consists of the pectoral girdle (scapula and clavicle on each side) and the upper limbs (humerus, radius, ulna, carpals, metacarpals, phalanges).

It also consists of the pelvic girdle (ilium, ischium and pubis on each side) and the lower limbs (femur, tibia, fibula, tarsals, metatarsals, phalanges, and the patella).

The girdles attach the limbs to the axial skeleton and provide surfaces for muscle attachment.

Q5(vi). Draw and label the human skull (described in words).

DIAGRAM DESCRIBED IN WORDS: the human skull has a cranium of 8 bones (frontal, 2 parietal, 2 temporal, occipital, sphenoid, ethmoid) enclosing the brain.

The face has 14 bones: 2 nasal, 2 maxillae, 2 zygomatic, 2 lacrimal, 2 palatine, 2 inferior nasal conchae, vomer and mandible.

The skull also contains the ear ossicles (malleus, incus, stapes) and the hyoid bone in the neck, and it protects the brain and sense organs.

Q5(vii). Write the major evolutionary adaptations in the lines of tetrapod.

Tetrapods evolved limbs positioned beneath the body, instead of sprawling at the sides, giving more efficient support and walking.

The limbs kept the pentadactyl (five-digit) plan, with girdles firmly attached to the vertebral column for strength.

Bones became longer and stronger, joints allowed new movements, and in mammals the spine arches during running to increase stride length.

Q5(viii). Define secondary growth. Explain.

Secondary growth is the increase in thickness (girth) of stems and roots of woody plants.

It is produced by lateral meristems: the vascular cambium, which makes secondary xylem (wood) inside and secondary phloem outside, and the cork cambium, which makes bark.

It differs from primary growth, which increases length and is produced by apical meristems.

Q5(ix). What are the main differences between exoskeleton and endoskeleton?

An exoskeleton is a hard covering outside the body, made of chitin or shell, while an endoskeleton is an internal framework of bone or cartilage.

An exoskeleton protects the body from outside but limits growth and must be moulted; an endoskeleton grows with the body and allows larger size.

An exoskeleton gives attachment to muscles on its inner surface, while an endoskeleton gives attachment on its outer surface.

Q5(x). List the functions of skeleton.

The skeleton supports the body and gives it shape, and protects soft organs such as the brain, heart and lungs.

It provides surfaces for muscle attachment and, with muscles, produces movement through levers at joints.

It stores calcium and phosphorus, and its red bone marrow produces blood cells.

Q5(xi). Explain the role of osteoclasts in remodelling of bone and describe the structure of compact bone.

Osteoclasts are large bone cells that break down and resorb old or damaged bone, releasing calcium into the blood.

Bone remodelling is the balanced work of osteoclasts (removal) and osteoblasts (formation), which repairs micro-damage and adjusts bone to stress.

Compact bone is made of osteons: concentric layers (lamellae) around a central Haversian canal carrying blood vessels and nerves, with osteocytes in lacunae connected by canaliculi.

Q5(xii). List the main parts of axial skeleton.

The axial skeleton consists of the skull (cranium and face), the vertebral column (cervical, thoracic, lumbar, sacral and coccygeal vertebrae), the ribs (12 pairs) and the sternum.

It forms the central axis of the body, protects the brain, spinal cord, heart and lungs, and supports the head and trunk.

The appendicular skeleton (limbs and girdles) attaches to this axial framework.

Q5(xiii). Distinguish between fibrous, cartilaginous and synovial joints.

Fibrous joints are held by dense fibrous tissue, allow no movement, and are found in skull sutures.

Cartilaginous joints are held by cartilage, allow slight movement, and are found between vertebrae and at the pubic symphysis.

Synovial joints have a fluid-filled cavity and allow free movement; examples are ball-and-socket (shoulder), hinge (elbow), pivot (neck) and gliding (wrist) joints.

Q5(xiv). Discuss methods of locomotion in fish, land vertebrates and birds.

Fish swim by S-shaped waves of muscle contraction passing along the body, with fins for steering and balance.

Land vertebrates walk, run or jump with tetrapod limbs placed under the body, using alternating limb movements and an arching spine.

Birds fly by flapping large wings, with a keeled sternum for flight-muscle attachment, light hollow bones, and feathers that form an aerofoil for lift.

Chapter 16 Q3 — flagged, not shown on the exercise pages.

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Chapter 17: Coordination and Control

Exercise Q1 — Fill in the blanks

Neurotransmitter molecules bind to the receptors on the postsynaptic membrane at a synapse.

Excess of ACTH hormone is secreted in Addison's disease.

Operant learning has been demonstrated and studied by Skinner and Thorndike .

Cytokinins are plant hormones which delay the life of fresh leaf crops.

All membranes of neurons have very active sodium and potassium pumps.

Exercise Q2 — True or False

Impulses travel much more rapidly along myelinated neurons.

True

All glial tissue consists of glial cells.

True

Saltatory conduction is carried out by those nerve fibres that have nodes of Ranvier.

True

The myelin sheath of a neuron is a particularly good conductor of electric impulse.

False

The resting membrane potential is maintained largely by the sodium pump.

True

Hormones initiate new biochemical reactions in the body.

False

Exercise Q4 — Short questions

Define circadian rhythm.

A circadian rhythm is a biological cycle of about 24 hours.

It controls daily patterns like sleep and wakefulness, hormone release and body temperature.

In humans it is set by the brain's clock and adjusted by daylight.

Jet lag occurs when travel disturbs this rhythm.

What is the difference between CNS and PNS?

The central nervous system consists of the brain and spinal cord.

It receives information, processes it and sends out commands.

The peripheral nervous system consists of nerves outside the brain and spinal cord.

It carries messages from sense organs to the CNS and from the CNS to muscles and glands.

What are the functions of the parathyroid gland?

The parathyroid glands secrete parathormone, which controls blood calcium level.

When calcium falls, parathormone releases calcium from bones into the blood.

It also increases calcium absorption from the gut and reduces its loss in urine.

This keeps blood calcium steady for nerves, muscles and blood clotting.

Define the term hormone.

A hormone is a chemical messenger secreted by an endocrine gland into the blood.

It travels to target cells elsewhere in the body and changes their activity.

Hormones act in very small amounts and do not start new reactions.

They speed up or slow down reactions already possible in the target cells.

What are the commercial applications of auxins?

Auxins are used to promote root formation in stem cuttings in nurseries.

They are sprayed to prevent premature fruit drop in orchards.

Synthetic auxins are used as selective weed killers for broad-leaved weeds.

They are also used to produce seedless fruits and to delay leaf fall.

List different types of tropisms.

Phototropism is bending toward or away from light, as in shoots growing toward

light.

Geotropism is growth in response to gravity: roots grow down and shoots grow up.

Thigmotropism is growth in response to touch, as in tendrils coiling around support.

Chemotropism is growth toward or away from chemicals, as in pollen tubes growing toward the ovule.

Exercise Q4 — Short questions (continued)

Q4(vii). Write a note on Alzheimer's disease.

Alzheimer's disease is a progressive brain disorder in which memory, thinking and behaviour slowly deteriorate.

It is associated with loss of brain cells, plaques of beta-amyloid protein and tangles of tau protein inside neurons, especially in the cortex and hippocampus.

There is no cure; treatment only relieves symptoms for a time, and care focuses on safety, routine and support for the patient.

Exercise Q5 — Extensive questions

Q5(i). Describe different types of learning behaviour.

Learning behaviour is behaviour modified by experience, and Thorpe classified it into six types.

Imprinting: young precocial birds follow the first moving object they see, normally the mother, during a brief sensitive period.

Habituation: the response to a repeated harmless stimulus decreases, for example a snail stops retracting when glass is tapped repeatedly.

Conditioning (type I): a neutral stimulus paired with a natural stimulus comes to produce the response, as Pavlov's dogs salivated at a bell.

Operant conditioning (type II): behaviour is shaped by reward and punishment through trial and error, studied by Thorndike and Skinner.

Latent learning: learning without obvious reward, shown when a rat that wandered a maze later runs straight to food.

Insight learning: solving a new problem by reasoning, as Kohler's chimpanzees stacked boxes to reach fruit.

Q5(ii). Describe in detail the role of adrenal glands.

The adrenal glands sit on top of the kidneys and have two parts with different roles.

The adrenal cortex secretes mineralocorticoids (aldosterone, controlling salt and water balance), glucocorticoids (cortisol, controlling glucose and stress response) and small amounts of sex hormones.

The adrenal medulla secretes adrenaline and noradrenaline, which prepare the body for emergency (fight or flight): faster heartbeat, higher blood sugar, wider airways and blood diverted to muscles.

Adrenal hormones are controlled by ACTH from the pituitary and by the sympathetic nervous system.

Q5(iii). Define nerve impulse. Explain the mechanism involved (diagrams described in words).

A nerve impulse is a wave of electrical depolarisation that travels along a neuron.

At rest, the membrane is polarised (about -70 mV inside) by the sodium-potassium pump, which keeps sodium outside and potassium inside.

When stimulated, sodium channels open, sodium rushes in and the membrane depolarises; then potassium channels open, potassium leaves and the membrane repolarises, followed by a brief refractory period.

In myelinated fibres the impulse jumps from node to node (saltatory conduction), which is much faster; the mechanism is illustrated by labelled diagrams in the book, described here in words.

Q5(iv). How is the nervous system of Planaria better developed than that of Hydra?

Hydra has a simple nerve net with no brain or central control, so responses are slow

and diffuse.

Planaria has a ladder-like nervous system: two longitudinal nerve cords joined by transverse nerves, with a bilobed brain (cerebral ganglia) at the head end.

Planaria can therefore show cephalisation, coordinated movement, and simple learning, which Hydra cannot.

Q5(v). Describe the structure and functions of the different parts of human brain.

The human brain has three main parts: forebrain, midbrain and hindbrain.

The forebrain contains the cerebrum (thinking, memory, senses, voluntary movement), thalamus (relay station) and hypothalamus (temperature, hunger, hormones).

The midbrain controls eye movements and relays visual and auditory signals.

The hindbrain contains the cerebellum (balance and coordination), pons (relay between cerebrum and cerebellum) and medulla (heartbeat, breathing, swallowing).

Q5(vi). Write a note on pituitary gland.

The pituitary gland, attached below the hypothalamus, is the master gland of the endocrine system.

Its anterior lobe secretes growth hormone, TSH, ACTH, FSH, LH and prolactin, controlling growth, thyroid, adrenals, gonads and milk production.

Its posterior lobe stores and releases ADH (water balance) and oxytocin (uterine contraction and milk release), both made in the hypothalamus.

Its middle lobe secretes MSH, which affects skin pigmentation.

Chapter 17 Q3 — flagged, not shown on the exercise pages.

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Chapter 18: Reproduction

Exercise Q1 — Fill in the blanks

Asexual reproduction requires only a single parent organism.

Sexual reproduction usually involves two parents.

Phytochromes are the special light sensitive pigments.

External fertilization occurs in aquatic environment.

Viviparous and ovoviviparous animals provide more protection to their young during development.

A placenta is established between the uterine and foetal tissues for the exchange of oxygen.

The reduction of progesterone level stimulates the pituitary gland to produce oxytocin hormone.

Exercise Q2 — True or False

Asexual reproduction involves mitotic cell division.

True

Asexually produced offspring are genetically identical to their parents.

True

Sexual reproduction involves a single parent.

False

Sexually produced offspring are identical to their parents.

False

Exercise Q4 — Short questions

What changes occur in ovulation and menstruation during pregnancy?

During pregnancy, ovulation stops because the corpus luteum keeps secreting

progesterone.

High progesterone prevents the ripening of new follicles in the ovary.

Menstruation also stops because the uterine lining is maintained for the embryo.

The lining is not shed, so no menstrual bleeding occurs until after birth.

What is the difference between oogenesis and spermatogenesis in humans?

Oogenesis produces one large ovum from each primary oocyte, with three polar bodies.

Spermatogenesis produces four small sperms from each primary spermatocyte.

Oogenesis begins before birth and pauses until puberty, while spermatogenesis starts at puberty.

Oogenesis is discontinuous with long rests, whereas spermatogenesis is continuous.

How is a seed formed?

After fertilization, the zygote develops into the embryo inside the ovule.

The ovule's integuments harden to form the seed coat.

The ovary wall develops into the fruit around the seed.

Stored food in the endosperm or cotyledons nourishes the embryo until germination.

What is the importance of seed in the life cycle of a plant?

The seed protects the embryo against drought, heat and cold.

It carries stored food, so the seedling can grow before it photosynthesises.

Seeds disperse to new places, spreading the species.

They also allow the plant to survive unfavourable seasons in a dormant state.

Exercise Q3 — Extensive questions

Q3(1). What structures are associated with the human female reproductive system? What are their functions?

The human female reproductive system consists of a pair of ovaries, fallopian tubes

(oviducts), uterus, cervix and vagina, with external genitalia.

The ovaries produce ova and the hormones oestrogen and progesterone.

The fallopian tubes carry the ovum from the ovary to the uterus and are the site of fertilisation.

The uterus receives and nourishes the developing embryo and foetus; the cervix is its lower opening into the vagina, which receives sperm and is the birth canal.

Q3(2). What are the functions of placenta during pregnancy?

During pregnancy the placenta nourishes the foetus: oxygen and nutrients pass from maternal blood to foetal blood, and carbon dioxide and wastes pass back, without the two bloods mixing.

It secretes hormones (progesterone, oestrogen, human chorionic gonadotrophin and placental lactogen) that maintain pregnancy and prepare the breasts for lactation.

It also acts as a selective barrier, filtering many harmful substances and microorganisms, although some drugs, alcohol and viruses can cross it.

Q3(3). Describe human menstrual cycle.

The menstrual cycle lasts about 28 days and is controlled by FSH, LH, oestrogen and progesterone.

Menstrual phase (days 1-5): the uterine lining sheds, causing bleeding.

Follicular phase (days 5-13): FSH stimulates a follicle to mature and oestrogen rebuilds the uterine lining.

Ovulation (about day 14): LH surge releases the ovum from the ovary.

Luteal phase (days 15-28): the corpus luteum secretes progesterone, which maintains the lining; if fertilisation does not occur, the corpus luteum degenerates and the cycle restarts.

Q3(4). Write notes on: (a) parthenogenesis, (b) herpes genitalia, (c) asexual reproduction, (d) seedless fruits.

(a) Parthenogenesis is development of an unfertilised egg into a new individual,

seen in some insects, rotifers and lizards, for example drones of honey bees.

(b) Genital herpes is a sexually transmitted infection caused by herpes simplex virus type 2, producing painful genital sores and ulcers; it can infect a baby during birth.

(c) Asexual reproduction is reproduction from a single parent without gametes, producing genetically identical offspring, for example budding in Hydra and binary fission in Amoeba.

(d) Seedless fruits develop without fertilisation (parthenocarpy), for example seedless bananas and grapes, often induced by auxins.

Chapter 19: Growth and Development

Exercise Q1 — Fill in the blanks

The influence of notochordal cells on the ectodermal cells to become the nervous system was called induction .

Microcephaly is a condition in which individuals have a small skull.

Growth is accompanied by two factors: (a) increase in cell number and (b) increase in cell size.

Meristems are the regions where growth is initiated by the proliferation of cells.

Exercise Q2 — True or False

Primary growth leads to increase in length, while secondary growth leads to increase in width.

True

The plants in which flowering is not at all affected by the day length are called day neutral plants.

True

The somatic mesoderm soon splits in the middle to form two layers: outer parietal layer and inner visceral layer.

True

In the clear cytoplasmic area, cytoplasm contains information essential for development.

True

The phase of cell movement and rearrangement is called cleavage.

False

Exercise — Short questions (selection)

Distinguish between primary and secondary growth in plants.

Primary growth increases the length of roots and shoots.

It is caused by apical meristems at the tips, which produce new cells.

Secondary growth increases the width (girth) of stems and roots.

It is caused by lateral meristems, the vascular cambium and cork cambium, in woody plants.

What is embryonic induction? Give an example.

Embryonic induction is the influence of one group of embryonic cells on neighbouring cells.

The inducing cells cause the neighbours to change their fate and develop into a particular tissue.

The classic example is the notochord inducing the overlying ectoderm to form the neural plate.

The neural plate then folds to form the brain and spinal cord.

What is the difference between growth and development?

Growth is an irreversible increase in size, mass or cell number of an organism.

Development is the whole series of changes from zygote to adult, including

differentiation.

Growth is quantitative, while development is both quantitative and qualitative.

An organism may stop growing but still continue to develop, as in metamorphosis.

Exercise Q4 — Short questions (continued)

Q4(i). What is organizer and inducer substance?

An organizer is a group of embryonic cells that directs the development of neighbouring cells, for example the dorsal lip of the blastopore in amphibians and the notochord.

An inducer substance is the chemical signal released by the organizer that causes neighbouring cells to change their fate.

For example, notochordal cells induce the overlying ectoderm to become the nervous system.

Q4(ii). What is differentiation?

Differentiation is the process by which unspecialised embryonic cells become specialised in structure and function.

During differentiation, cells switch on only some of their genes and develop particular shapes and functions, for example nerve cells, muscle cells and blood cells.

Once differentiated, a cell usually keeps its special character and divides to give more cells of the same type.

Q4(v). What is meristem?

A meristem is a region of actively dividing, undifferentiated cells in a plant, where growth is initiated.

Apical meristems at root and shoot tips cause increase in length (primary growth).

Lateral meristems (cambium) cause increase in thickness (secondary growth), and intercalary meristems occur at the bases of leaves and internodes in grasses.

Exercise Q5 — Extensive questions

Q5(i). What is aging? How will you explain this process?

Aging (senescence) is the progressive decline in the body's structure and function with time.

It can be explained by wear and tear of cells, accumulation of mutations and damage by free radicals, shortening of telomeres at each cell division (the genetic clock), and declining hormone production.

No single cause explains aging; genetic programming and environmental damage act together.

Q5(ii). What is regeneration? Why is it so effective in some animals and missing in others?

Regeneration is the ability to regrow a lost or injured body part.

It is very effective in simple animals such as sponges, planaria and starfish, because they have unspecialised cells (neoblasts) that can migrate and differentiate, or their cells can dedifferentiate and form new tissues.

It is missing or weak in higher animals because their cells are highly specialised, wounds heal by scar tissue instead of regrowth, and complex organs cannot be rebuilt.

Q5(iii). Describe in detail the developmental processes of chick.

The chick develops from a fertilised egg in about 21 days of incubation.

Cleavage produces a blastoderm on the yolk, then gastrulation through the primitive streak forms ectoderm, mesoderm and endoderm.

Neurulation forms the neural tube, and organogenesis builds the heart, brain, limbs and other organs in order.

Extra-embryonic membranes protect and feed the embryo: amnion (fluid), chorion (gas exchange), allantois (waste storage and respiration) and yolk sac (food), until the chick hatches.

Q5(iv). What is growth? Discuss different phases and conditions for growth.

Growth is an irreversible increase in size, mass or cell number of an organism.

Its phases are: formative phase (rapid cell division), elongation phase (cells enlarge by taking up water), and maturation phase (cells differentiate and reach full size).

Conditions for growth are: adequate nutrients and water, suitable temperature, oxygen for respiration, light (for plants), and hormones such as auxins, gibberellins and growth hormone.

Q5(v). What is development? Describe the principles of development in detail.

Development is the whole series of orderly changes by which a single cell becomes a complete organism.

Its principles are: cell division increases cell number; growth increases size; differentiation makes cells specialised; morphogenesis shapes organs; and induction directs neighbouring cells to their fates.

Development is also controlled by gradients, polarity of the egg, programmed cell death (apoptosis) that removes temporary structures, and interaction with the environment.

Chapter 19 Q3 — flagged, not shown on the exercise pages.

FLAGGED — BOOK NUMBERING NOTE: the Chapter 19 exercise pages as extracted show Q1, Q2, Q4 and Q5; no Q3 appears on those pages.

Q3 is therefore not guessed here; this is a render-verified book numbering skip in the book.

Chapter 20: Chromosomes and DNA

Exercise Q1 — Fill in the blanks

Particular tRNA molecules become attached to specific amino acids through the

action of activating enzymes called aminoacyl-tRNA synthetases .

Transformation is the transfer of genetic material from one cell to another and can alter the genetic make-up of the recipient cell.

In a bacterium, a subunit of RNA polymerase called sigma factor recognizes the -10 sequence in the promoter and binds RNA polymerase there.

A typical human chromosome contains about 140 million nucleotides in its DNA.

Miescher extracted a white substance from the nuclei of human cells and fish sperm and called this substance nuclein .

Exercise Q2 — True or False

The strand of DNA that is not transcribed is called the coding strand.

True

The TATAAT sequence called the -35 sequence is part of the promoter, where transcription actually starts.

False

Rosalind Franklin carried out an X-ray diffraction analysis of DNA.

True

The base pairs in the DNA helix are planar and stack 34 nm apart as a result of hydrophobic interactions.

False

Exercise Q4 — Short questions

What are the three major classes of RNA?

Messenger RNA (mRNA) carries the genetic code from DNA to the ribosome.

Transfer RNA (tRNA) brings amino acids to the ribosome and reads the codons.

Ribosomal RNA (rRNA) forms the structure of ribosomes, where proteins are made.

Together the three RNAs carry out protein synthesis in the cell.

What is the function of RNA polymerase in transcription?

RNA polymerase binds to the promoter region of a gene on DNA.

It unwinds the DNA double helix to expose the template strand.

It then joins ribonucleotides complementary to the template to build the RNA chain.

It stops at the terminator sequence and releases the completed RNA molecule.

What is an anticodon?

An anticodon is a sequence of three bases on a tRNA molecule.

It is complementary to a codon on the mRNA during translation.

Each tRNA carries the amino acid specified by the codon it matches.

Anticodon-codon pairing ensures amino acids are added in the correct order.

How did Crick and his colleagues determine how many nucleotides specify each amino acid?

They used frameshift mutations in bacteriophages, adding or deleting single bases.

One added or deleted base shifted the reading frame and destroyed protein function.

Three added or deleted bases together restored the frame and the function.

This proved that the code is read in groups of three nucleotides, or triplets.

Exercise Q5 — Extensive questions

Q5(1). How did Hershey and Chase determine which component of bacterial viruses contains the hereditary information?

Hershey and Chase worked with T2 bacteriophages, which infect bacteria.

They labelled the phage DNA with radioactive phosphorus (32P) and the phage protein coat with radioactive sulphur (35S), then let the phages infect bacteria.

After shaking off the coats and centrifuging, the 32P (DNA) was found inside the bacterial cells and the 35S (protein) outside.

This proved that DNA, not protein, enters the cell and carries the hereditary

information.

Q5(2). What is the three dimensional shape of DNA? How does it fit with Chargaff's observations on purines and pyrimidines?

DNA is a double helix: two antiparallel polynucleotide strands wound around each other, with sugar-phosphate backbones outside and bases inside.

Adenine pairs with thymine by two hydrogen bonds, and guanine pairs with cytosine by three hydrogen bonds.

This fits Chargaff's observations: in any DNA, adenine equals thymine and guanine equals cytosine, and total purines equal total pyrimidines, exactly as base pairing predicts.

Q5(3). How did Meselson and Stahl show that DNA replication is semiconservative?

Meselson and Stahl grew E. coli for many generations in heavy nitrogen (15N), so all DNA was heavy, then transferred the bacteria to light nitrogen (14N).

After one generation in 14N, centrifugation showed a single hybrid-density band, as semiconservative replication predicts (each molecule half old, half new).

After two generations, two bands appeared: one hybrid and one light, which ruled out conservative replication and confirmed semiconservative replication.

Q5(4). What is the basis for the requirement that leading and lagging strands be replicated by different mechanisms?

DNA polymerase can only add nucleotides in the 5' to 3' direction, and the two strands of DNA are antiparallel.

On one template strand, synthesis runs continuously toward the replication fork: this is the leading strand.

On the other template strand, synthesis must run away from the fork in short Okazaki fragments, which DNA ligase joins: this is the lagging strand.

The different mechanisms are therefore forced by the antiparallel structure and the one-way action of DNA polymerase.

Q5(5). What hypothesis did Beadle and Tatum test in their experiments on Neurospora?

Beadle and Tatum tested the hypothesis that one gene controls one enzyme in a metabolic pathway.

They irradiated Neurospora (bread mould) to create mutants, then grew each mutant on minimal medium and on media supplemented with single nutrients.

A mutant that grew only when a particular nutrient was added was blocked at one step of a pathway, showing that one gene had been damaged and one enzyme was missing.

This established the one gene-one enzyme (later one gene-one polypeptide) hypothesis.

Chapter 20 Q3 — flagged, not shown on the exercise pages.

FLAGGED — BOOK NUMBERING NOTE: the Chapter 20 exercise pages as extracted show Q1, Q2, Q4 and Q5; no Q3 appears on those pages.

Q3 is therefore not guessed here; this is a render-verified book numbering skip in the book.

Chapter 21: Cell Cycle

Exercise Q1 — Fill in the blanks

Mongolism is also known as Down syndrome .

During zygotene homologous chromosomes get close to each other.

S phase precedes G2 phase.

Polar microtubules elongate during anaphase.

The mitotic apparatus is formed during prophase of cell division.

The chromosome number (44+1) denotes Turner syndrome.

Intracellular contents are released during the type of cell death called necrosis .

Exercise Q3 — True or False

Meiosis occurs in haploid cells only.

False

Cell cycle is comprised of two phases, i.e., karyokinesis and cytokinesis.

False

A point where non-sister chromatids cross each other is called kinetochore.

False

G0 stands for no gap.

False

Full life cycle of yeast cells requires 90 seconds to be completed.

False

Crossing over takes place during metaphase I.

False

Autosomal non-disjunction may occur in chromosomes other than the 21st chromosome.

True

Benign tumours are always non-localized.

False

Cancer is caused mainly by mutations in germ cells.

False

Genetic information remains unchanged during mitosis.

True

Homologous chromosomes are necessarily identical.

False

The cells are kept alive due to trophic factors.

True

Cytokinesis involves the division of cytochromes.

False

Phragmoplast is a type of fragmentation.

False

Exercise Q4 — Short questions

Differentiate between necrosis and apoptosis.

Necrosis is accidental cell death caused by injury, toxins or lack of blood supply.

The cell swells, bursts and releases its contents, causing inflammation in nearby tissue.

Apoptosis is programmed, orderly cell death controlled by the cell's own genes.

The cell shrinks into neat bodies that are eaten by phagocytes, without inflammation.

What are the functions of the mitotic apparatus?

The mitotic apparatus consists of spindle fibres, asters and centrioles in animal cells.

It attaches to the centromeres of chromosomes and lines them up at the equator.

During anaphase it pulls sister chromatids apart to opposite poles.

It therefore ensures that each daughter cell receives an equal set of chromosomes.

Give the importance and significance of meiosis.

Meiosis halves the chromosome number, so fertilization restores the diploid number.

Crossing over and independent assortment create new gene combinations.

This variation is the raw material for evolution and adaptation.

Meiosis also repairs genetic damage and produces haploid gametes for sexual reproduction.

Define chromosomal non-disjunction.

Non-disjunction is the failure of homologous chromosomes or sister chromatids to separate.

It occurs during anaphase of meiosis I or meiosis II.

The result is gametes with an extra chromosome or missing a chromosome.

Fertilization with such a gamete causes syndromes like Down syndrome or Turner syndrome.

What are the symptoms of Turner syndrome?

Turner syndrome occurs in females with only one X chromosome (44+X).

Affected girls are short in stature with a webbed neck and a broad chest.

Their ovaries do not develop, so they are infertile and do not menstruate.

They may also have heart defects and learning difficulties.

Define cell cycle. Highlight its importance and significance.

The cell cycle is the ordered series of events from one cell division to the next.

It consists of interphase (G1, S, G2) and the mitotic phase (mitosis and cytokinesis).

It ensures accurate copying of DNA and equal distribution to daughter cells.

Controlled cell cycles allow growth, repair and replacement of worn-out cells.

Is interphase a resting phase? Why?

No, interphase is not a resting phase, although the cell is not visibly dividing.

It is the longest and most active phase of the cell cycle.

During G1 the cell grows and makes proteins; during S it copies its entire DNA.

During G2 it prepares the machinery needed for mitosis, so it is a period of intense activity.

In what respect does mitosis in plant cells differ from that in animal cells?

Plant cells have no centrioles, so the spindle forms without asters.

Animal cells divide by a cleavage furrow that pinches the cell in two.

Plant cells cannot furrow because of the rigid cell wall, so they build a cell plate.

The cell plate grows outward and becomes the new cell wall between daughter

cells.

Exercise Q4 — Short questions (continued)

Q4(3). How can you identify the cancer cells?

Cancer cells can be identified by uncontrolled, rapid division that does not stop at the normal limits of the tissue.

They lose contact inhibition, so they pile up instead of stopping when they touch neighbouring cells, and they have large, irregular nuclei and abnormal chromosome numbers.

They do not differentiate normally, they invade surrounding tissues, and malignant cells spread (metastasise) through blood and lymph to form tumours elsewhere.

Exercise Q3 — Extensive questions

Q3(1). How does cytokinesis occur in animal cells? In which way does it differ from that in plant cells?

In animal cells, cytokinesis occurs by a cleavage furrow: a contractile ring of actin and myosin pinches the cell membrane inward until the cell divides into two.

In plant cells, the rigid wall prevents pinching, so vesicles from the Golgi apparatus gather at the centre (the phragmoplast) and fuse to form a cell plate.

The cell plate grows outward to the existing walls and becomes the new cell wall and middle lamella between the two daughter cells.

Q3(2). Why and how do the chromosomes get separated during anaphase of mitosis?

During anaphase of mitosis, the sister chromatids are pulled apart to opposite poles of the cell.

Kinetochore microtubules attached to the centromeres shorten, drawing each chromatid toward its pole, while polar microtubules elongate and push the poles

apart.

This separation is driven by motor proteins and by controlled breakdown of microtubules, so each daughter cell receives one copy of every chromosome.

Q3(3). What is the role of centriole in an animal cell? How is this function carried out in plant cells?

In an animal cell, the centriole pair forms the centrosome, which organises the spindle: microtubules grow from the centrosomes and attach to chromosomes. Centrioles also form the basal bodies of cilia and flagella.

Plant cells have no centrioles; their spindle is organised by microtubule-organising centres at the poles, and the spindle forms without asters.

Q3(4). In what respect can cell death be regarded beneficial?

Cell death can be beneficial because it shapes the body and removes what is no longer needed.

Apoptosis removes the tissue between developing digits, the tail of a tadpole during metamorphosis, and the large excess of neurons produced during brain development.

It also removes damaged, infected or potentially cancerous cells without releasing their contents, so neighbouring cells are not harmed.

Q3(5). Compare mitosis with meiosis and describe their importance.

Mitosis is one division producing two genetically identical diploid cells; it is important for growth, repair and asexual reproduction.

Meiosis is two divisions producing four genetically different haploid cells; it is important for sexual reproduction, keeping the chromosome number constant, and creating variation.

Mitosis conserves the chromosome number and genotype, while meiosis halves the number and shuffles genes by crossing over and independent assortment.

Q3(6). Define disjunction and discuss its effect.

Disjunction is the normal separation of homologous chromosomes (meiosis I) or sister chromatids (meiosis II and mitosis) to opposite poles.

Non-disjunction is the failure of this separation, so a gamete receives an extra chromosome or loses one.

Its effects are aneuploid offspring: Down syndrome (extra chromosome 21), Turner syndrome (one X only) and Klinefelter syndrome (XXY) are examples.

Q3(7). Describe meiosis and explain its significance.

Meiosis is the division that produces haploid gametes from a diploid cell in two successive divisions, meiosis I and meiosis II.

In meiosis I, homologous chromosomes pair, exchange segments by crossing over, and separate, halving the chromosome number.

In meiosis II, sister chromatids separate, giving four haploid cells.

Its significance is that it keeps the chromosome number constant across generations and creates genetic variation through crossing over and independent assortment.

Chapter 21 Q2 — flagged, not shown on the exercise pages.

FLAGGED — BOOK NUMBERING NOTE: the Chapter 21 exercise pages as extracted show Q1, Q3 (True/False) and Q4 (short questions) and Q3 (extensive questions, printed after Q4); no Q2 appears on those pages.

Q2 is therefore not guessed here; this is a render-verified book numbering skip in the book.

Chapter 22: Variation and Genetics

Exercise Q1 — Fill in the blanks

Gene is the basic unit of biological information.

A sudden change in the structure of a gene is called mutation .

Probability is the chance of an event to occur.

A cross among monohybrids is a monohybrid cross.

An individual with a homozygous genotype is called pure .

Different alleles of a gene that are both expressed in a heterozygote are called codominant alleles.

When a heterozygote exceeds the phenotypic expression of both the homozygotes, the phenomenon is called overdominance .

When a single gene affects two or more traits, the phenomenon is called pleiotropy .

A gene with multiple phenotypic effects is called a pleiotropic gene .

The phenomenon of staying together of all the genes of a chromosome is called linkage .

Linkage minimizes the chances of genetic recombination.

Crossing over is an exchange of segments between non-sister chromatids of homologous chromosomes during meiosis.

All chromosomes other than sex chromosomes are called autosomes .

SRY gene is the maleness determining gene in man.

Type II diabetes mellitus is non-insulin dependent.

Polygenic inheritance with environmental influence is called multifactorial inheritance.

Exercise Q2 — True or False

In grasshopper, the male has XY and the female has XX types of sex chromosomes.

False

Pea is normally a self-fertilizing plant.

True

Dihybrids are offspring of the parents who differ in one contrasting pair of trait.

False

X-linked traits pass directly from father to son.

False

A person suffering from blue cone monochromacy cannot see blue colour.

False

In birds and moths, eggs determine sex.

True

A homozygote forms all gametes of the same type.

True

The allele for a sex-limited trait is dominant in one sex but recessive in the other.

False

Exercise — Short questions (selection)

State Mendel's law of segregation with an example.

The law of segregation states that the two alleles of a gene separate during gamete formation.

Each gamete receives only one allele, so offspring receive one allele from each parent.

In pea plants, a cross between pure tall (TT) and pure dwarf (tt) gives all tall F1 plants.

In F2, the alleles segregate again to give tall and dwarf plants in a 3:1 ratio.

What is the difference between genotype and phenotype?

Genotype is the genetic make-up of an organism, the alleles it carries.

Phenotype is the observable character produced by the genotype and the environment.

Two organisms can have the same phenotype but different genotypes, like TT and Tt

tall peas.

Phenotype can change with environment, but genotype remains fixed.

Exercise Q2 — True or False (continued: items 9-10)

Pattern baldness is a sex influenced trait.

True

Carriers of haemophilia show no symptoms of the disease.

True

Exercise Q4 — Short questions

Q4(1). Differentiate between the pairs listed in the book (phenotype/genotype, gene/allele, homozygous/heterozygous, monohybrid/dihybrid, autosome/sex chromosome, dominance/epistasis, allele/multiple allele, X-linked/Y-linked, incomplete dominance/codominance, sex-limited/sex-influenced, continuous/discontinuous variation, dominant/recessive trait, wild type/mutant).

(1) Phenotype is the observable character; genotype is the genetic make-up (alleles) behind it. Gene is a unit of inheritance at a locus; allele is one alternative form of that gene.

Homozygous means two identical alleles; heterozygous means two different alleles.

Monohybrid cross involves one contrasting pair; dihybrid cross involves two pairs.

Autosomes are non-sex chromosomes; sex chromosomes determine sex.

Dominance is masking of one allele by another; epistasis is masking of one gene by another gene.

Allele is one form of a gene; multiple alleles are three or more forms in a population (for example ABO). X-linked traits are carried on the X chromosome; Y-linked traits pass only from father to son.

Incomplete dominance gives an intermediate heterozygote (pink snapdragon);

codominance expresses both alleles (AB blood group). Sex-limited traits appear in one sex only (milk yield); sex-influenced traits appear in both sexes but more in one (pattern baldness).

Continuous variation is graded (height); discontinuous variation is distinct classes (blood group). Dominant trait appears in the heterozygote; recessive trait appears only in the homozygote. Wild type is the common normal form; mutant is the changed form.

(Note: the genotype/phenotype pair is already covered in the existing notes and is included here only as part of the book's Q4(1) list.)

Q4(2). What is a gene pool?

A gene pool is the total aggregate of all the genes (alleles) in a population at a given time.

It includes every allele carried by every individual, and evolution is a change in the frequencies of these alleles over generations.

Q4(3). Was pea a lucky choice for Mendel? What would have happened if he had studied an eighth character?

Yes, pea was a lucky choice: it is easy to grow, has a short generation time, produces many offspring, self-fertilises but can also be cross-pollinated, and has clear contrasting characters.

Mendel's seven characters mostly lay on different chromosomes, so they assorted independently.

If he had studied an eighth character that was linked to one of the seven (on the same chromosome), the two characters would not have assorted independently, and his simple ratios would have been disturbed.

Q4(4). What is a test cross? Why did Mendel devise this cross?

A test cross is a cross of an individual of unknown genotype with a homozygous recessive individual.

Mendel devised it to reveal the unknown genotype: if all offspring show the dominant trait, the parent is homozygous dominant; if about half show the recessive trait, the parent is heterozygous.

It is still used to test whether a dominant individual is pure or hybrid.

Q4(5). What would happen if alleles of a pair do not segregate at meiosis? How would it affect the purity of gamete?

If alleles of a pair do not segregate at meiosis (non-disjunction), some gametes receive both alleles and others receive neither.

The purity of gametes is lost: gametes are no longer haploid for that chromosome, and the offspring receive an abnormal chromosome number (aneuploidy).

This can cause syndromes such as Down syndrome when chromosome 21 fails to separate.

Q4(6). If the alleles do not assort independently, which type of combination is missing in the progeny?

If alleles do not assort independently, the recombinant (new) combinations are missing in the progeny.

Only the parental combinations appear, because the genes stay together on the same chromosome (linkage) instead of shuffling freely.

Recombinant types appear only if crossing over separates the linked genes.

Q4(7). Why does each gamete have an equal chance of getting one or the other allele of a pair?

During meiosis I, the two homologous chromosomes of a pair separate, and each gamete receives one of them.

Which homologue goes to a given gamete is random, so each gamete has an equal chance of receiving either allele of the pair.

This random separation is the physical basis of Mendel's law of segregation.

Q4(8). Does the dominant allele modify the determinative nature of its recessive

partner? What sort of relationship do they have?

No, the dominant allele does not modify or destroy the recessive allele; the recessive allele remains unchanged in the heterozygote and reappears in later generations.

Their relationship is dominance: the dominant allele's product masks the recessive allele's effect in the heterozygote, while the recessive allele is expressed only when homozygous.

This is not blending inheritance, because the two alleles keep their separate identities.

Q4(9). Which type of traits can assort independently?

Traits whose genes lie on different chromosomes assort independently, because the chromosomes themselves assort independently at meiosis.

Genes far apart on the same chromosome can also assort almost independently, because crossing over between them is frequent.

Closely linked genes on the same chromosome do not assort independently.

Q4(10). Why does the blood group phenotype of a person remain constant throughout life?

A person's blood group phenotype remains constant because it is determined by the genotype (the ABO and Rh genes) inherited at conception.

The antigens on the red blood cells are produced throughout life from those genes and do not change with age, diet or disease.

Only a bone-marrow transplant, which replaces the blood-forming cells, could change the blood group.

Q4(11). What is a universal blood donor?

A universal blood donor is a person of blood group O negative, whose red blood cells carry neither A nor B antigens and no Rh antigen.

Such blood can be given to recipients of any ABO group without the recipient's

antibodies attacking the donor cells, which is vital in emergencies.

Universal donors can themselves receive only O negative blood.

Q4(12). How can you protect the baby against Rh-incompatibility?

The baby is protected by giving the Rh-negative mother an anti-D (anti-Rh) immunoglobulin injection, usually after the first delivery (and after any miscarriage or abortion), within 72 hours.

The anti-D antibodies destroy any Rh-positive foetal cells that entered the mother's blood before her immune system can make its own antibodies.

This prevents sensitisation, so a later Rh-positive baby is not attacked by maternal antibodies (erythroblastosis foetalis).

Q4(13). Which type of genes do not obey law of independent assortment?

Genes that lie on the same chromosome do not obey the law of independent assortment, because they tend to be inherited together (linkage).

Mendel's law holds only for genes on different chromosomes, or far apart on the same chromosome where crossing over is frequent.

Linked genes are separated only by crossing over during meiosis.

Q4(14). How can linked genes be separated from each other?

Linked genes are separated from each other by crossing over between non-sister chromatids of homologous chromosomes during prophase I of meiosis.

The farther apart two genes are on a chromosome, the more likely crossing over occurs between them, and the higher the recombination frequency.

This is why recombination frequency is used to map the relative positions of genes on a chromosome.

Q4(15). What is multifactorial inheritance?

Multifactorial inheritance is inheritance controlled by many genes (polygenes) together with environmental influence.

Examples are height, skin colour, blood pressure and diabetes mellitus type II,

where the trait varies continuously and is affected by diet, exercise and other factors.

It differs from single-gene inheritance, where one gene gives distinct classes.

Q4(16). What is MODY?

MODY is maturity-onset diabetes of the young: diabetes that appears early in life, usually before 25 years of age, in non-obese people.

It is inherited as an autosomal dominant trait and is caused by mutations in genes controlling insulin secretion, most commonly the glucokinase gene (about 50% of cases).

Unlike type II diabetes of older adults, MODY is a single-gene disorder, though several different MODY genes are known.

Q4(17). Can a child have more intelligence (IQ score) than his parents?

Yes, a child can have a higher IQ score than both parents, because intelligence is a polygenic, multifactorial trait.

A child inherits a new combination of many alleles from both parents, and recombination can bring together more favourable alleles than either parent carries.

Environment also matters: nutrition, education, health and stimulation can raise or lower the expressed IQ, so children are not limited to their parents' scores.

Exercise Q4 — Extensive questions (book's second Q4 set)

Q4-Ext(1). What is incomplete dominance? Explain it with an example.

Incomplete dominance is inheritance in which the heterozygote shows a phenotype intermediate between the two homozygotes, because neither allele is fully dominant.

In snapdragons, red (RR) crossed with white (rr) gives pink (Rr) flowers in the F1, and the F2 ratio is 1 red : 2 pink : 1 white, the same for genotype and phenotype.

Q4-Ext(3). Define Mendel's law of independent assortment. Explain it with an

example.

Mendel's law of independent assortment states that when two or more pairs of contrasting characters are combined, the alleles of one pair assort independently of the alleles of another pair during gamete formation.

For example, in a dihybrid cross of pea (round yellow x wrinkled green), the F2 generation gives 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green.

This happens because the genes for seed shape and seed colour lie on different chromosomes and separate independently at meiosis.

Q4-Ext(4). Define probability. Derive the 9:3:3:1 F2 ratio of independent assortment through the product rule.

Probability is the chance that an event occurs, calculated as the number of favourable outcomes divided by the total number of outcomes.

In a dihybrid cross, the probability of round seeds is 3/4 and of yellow seeds is 3/4; by the product rule, the probability of round yellow seeds is 3/4 x 3/4 = 9/16.

Applying the product rule to all four combinations gives 9/16 round yellow, 3/16 round green, 3/16 wrinkled yellow and 1/16 wrinkled green: the 9:3:3:1 ratio.

Q4-Ext(5). What is codominance? Explain the phenomenon of codominance with an example.

Codominance is inheritance in which both alleles of a gene are fully expressed in the heterozygote, with neither masking the other.

In human ABO blood groups, a person with genotype IAIB has blood group AB, because both A and B antigens are present on the red blood cells.

In roan cattle, red and white alleles together give a coat with both red and white hairs.

Q4-Ext(6). Define multiple alleles. Describe the multiple allelic blood group system of man.

Multiple alleles are three or more alternative forms of a gene present in a population, although any one individual carries only two.

The human ABO blood group system has three alleles: IA, IB and i. IA and IB are codominant with each other and dominant over i.

The six genotypes give four phenotypes: A (IAIA, IAi), B (IBIB, IBi), AB (IAIB) and O (ii).

Q4-Ext(7). What is Rh factor? Describe the genetic basis of Rh blood group system of man.

The Rh factor is an antigen (D antigen) on red blood cells, controlled by a dominant gene; people with it are Rh-positive, people without it are Rh-negative.

The Rh system is inherited as a single-gene trait: Rh-positive may be homozygous (DD) or heterozygous (Dd), while Rh-negative is homozygous recessive (dd).

An Rh-negative mother carrying an Rh-positive baby may become sensitised at delivery, which endangers later Rh-positive babies.

Q4-Ext(8). What is erythroblastosis foetalis? Discuss its adverse effect and therapy to avoid Rh sensitisation.

Erythroblastosis foetalis is a haemolytic disease of the newborn caused by Rh incompatibility.

An Rh-negative mother sensitised by an earlier Rh-positive baby makes anti-Rh antibodies, which cross the placenta in a later Rh-positive pregnancy and destroy the baby's red blood cells, causing anaemia and jaundice.

It is prevented by giving the mother anti-D immunoglobulin after delivery, and treated in severe cases by exchange transfusion of the baby.

Q4-Ext(9). Define epistasis. Explain epistatic gene interaction with an example.

Epistasis is the interaction in which one gene masks or modifies the expression of another gene at a different locus.

In mice, coat colour is controlled by a colour gene (B black, b brown) and an

epistatic gene (C allows colour, cc gives albino whatever the colour gene).

A dihybrid cross gives a 9:3:4 ratio (9 black : 3 brown : 4 albino), instead of 9:3:3:1, because the cc genotype hides the colour gene.

Q4-Ext(10). What is a pleiotropic gene? Discuss pleiotropy with examples.

A pleiotropic gene is a single gene that affects two or more apparently unrelated traits.

Sickle-cell anaemia is pleiotropic: one mutation in the haemoglobin gene causes sickled red cells, anaemia, organ damage and resistance to malaria.

Phenylketonuria is also pleiotropic: failure to break down phenylalanine causes mental retardation, light skin and a musty body odour.

Q4-Ext(11). What are polygenes? Explain polygenic inheritance.

Polygenes are many genes, each with a small effect, that together control a single quantitative character.

Polygenic inheritance gives continuous variation: for example, human skin colour is controlled by several genes, producing a graded range from light to dark rather than two classes.

Height, weight and intelligence are also polygenic, and are influenced by the environment as well.

Q4-Ext(12). What is crossing over? Define recombination frequency and explain its significance.

Crossing over is the exchange of segments between non-sister chromatids of homologous chromosomes during prophase I of meiosis.

Recombination frequency is the percentage of recombinant offspring among the total, and it measures how often crossing over separates two genes.

Its significance is that it shows whether genes are linked and how far apart they lie: a higher frequency means the genes are farther apart, which allows gene mapping.

Q4-Ext(13). What are sex-chromosomes? Discuss the chromosomal patterns of sex

determination in organisms.

Sex chromosomes are the chromosomes that determine the sex of an individual; the rest are autosomes.

In humans and Drosophila, females are XX and males XY; in birds and moths, males are ZZ and females ZW (eggs determine sex); in grasshoppers, females are XX and males XO.

The pattern differs between groups, but in each case the heterogametic sex produces two types of gametes and determines the sex of the offspring.

Q4-Ext(14). Compare chromosomal determination of sex between Drosophila and humans.

In humans, sex is determined by the Y chromosome: the SRY gene on the Y triggers male development, so XY individuals are male and XX are female.

In Drosophila, sex is determined by the ratio of X chromosomes to sets of autosomes: a ratio of 1.0 gives a female, 0.5 gives a male, and the Y chromosome is needed for male fertility but not for maleness itself.

Thus the same XX/XY system works by different mechanisms in the two organisms.

Q4-Ext(15). Define gene pool. Explain the concept of gene pool in a sample population.

A gene pool is the total aggregate of genes in a population at any time.

In a sample population, the gene pool can be described by allele frequencies: for example, if 60% of the alleles at a locus are A and 40% are a, those frequencies define the pool.

Evolution is a change in these frequencies over generations, caused by mutation, selection, migration, drift and non-random mating.

Q4-Ext(16). What is sex linkage? Explain T. H. Morgan's study of sex-linkage in Drosophila.

Sex linkage is inheritance of genes carried on the sex chromosomes, usually the X

chromosome.

Morgan studied white eye in Drosophila: a white-eyed male crossed with a red-eyed female gave all red-eyed F1, but in the F2 all white-eyed flies were male.

This showed the eye-colour gene lies on the X chromosome: males, with one X, express the recessive allele, while females need two copies.

Q4-Ext(17). Compare the pattern of inheritance of an X-linked dominant trait with an X-linked recessive trait in humans.

An X-linked recessive trait (for example haemophilia) appears mostly in males, passes from carrier mothers to sons, and never passes directly from father to son.

An X-linked dominant trait appears in both sexes, but an affected father passes it to all his daughters and none of his sons, while an affected heterozygous mother passes it to half her children of either sex.

Recessive traits can skip generations through carriers; dominant traits usually appear in every generation.

Q4-Ext(18). Explain diabetes mellitus and its genetic basis.

Diabetes mellitus is a group of disorders with high blood sugar, because the body cannot properly use glucose.

Type I (insulin-dependent) is an autoimmune destruction of pancreatic beta cells, so no insulin is made; it usually starts young and needs insulin injections.

Type II (non-insulin-dependent, about 90% of cases) is insulin resistance, often with obesity and age over 40; it is multifactorial, with several genes (including the insulin gene on chromosome 11) and environmental influence.

MODY is an early-onset autosomal dominant form, often caused by glucokinase gene mutations.

Q4-Ext(19). Discuss the genetics of colour-blindness or haemophilia.

Colour blindness (red-green) is an X-linked recessive trait: affected males have only one X carrying the faulty allele, while females need two copies.

A colour-blind father passes the allele to all his daughters (who become carriers) and to none of his sons.

Haemophilia is also X-linked recessive: the blood fails to clot because a clotting factor is missing, so affected males bleed excessively from small wounds, and carrier females usually show no symptoms.

Chapter 22 Q3 — flagged, not shown on the exercise pages.

NOTE: extensive item (2) (Mendel's law of segregation) is already covered in the existing notes and is not duplicated here.

FLAGGED — render-verified book numbering skip: the Chapter 22 exercise pages as extracted show Q1, Q2 and Q4; no Q3 appears on those pages.

Chapter 23: Biotechnology

Exercise Q1 — Fill in the blanks

The use of polymerase chain reaction (PCR) creates millions of copies in a laboratory test tube.

Transgenic free-living organisms in the environment have had a foreign gene inserted into them.

Probes are known sequences of DNA that are used to find complementary DNA strands; they can be used diagnostically to determine the presence of a particular gene.

Exercise Q3/Q4 — Short and extensive questions

How and why are transgenic animals that secrete a product often cloned?

A transgenic animal is first produced by inserting a useful gene into its genome. Such animals are rare and valuable, so they are cloned to make many identical

copies.

Cloning is done by nuclear transfer into an egg cell, which develops into an embryo.

All the clones then secrete the same product, like a medicine, in their milk.

Explain two primary goals of the Human Genome Project. What are the possible benefits of the project?

The first goal was to determine the complete sequence of all three billion base pairs in human DNA.

The second goal was to identify all human genes and map them on their chromosomes.

Benefits include diagnosis and treatment of genetic diseases before symptoms appear.

It also helps in making new medicines and understanding human evolution.

Explain and give examples of ex vivo and in vivo gene therapies in humans.

In ex vivo gene therapy, cells are removed from the patient, corrected in the laboratory and returned.

It was used to treat severe combined immunodeficiency (SCID) by correcting blood stem cells.

In in vivo gene therapy, the corrected gene is delivered directly into the patient's body.

It is used for cystic fibrosis, where a working gene is delivered to lung cells by a vector.

What is the polymerase chain reaction (PCR), and how is it carried out to produce multiple copies of a DNA segment?

PCR is a laboratory method that makes millions of copies of a chosen DNA segment.

First, the DNA is heated to separate its two strands.

Primers bind to the target sequence and DNA polymerase builds new strands.

Repeating these cycles about thirty times produces over a billion copies.

What is DNA fingerprinting, a process that utilizes the entire genome?

DNA fingerprinting compares patterns of DNA from different individuals.

DNA is cut into fragments, separated by size and probed for variable regions.

Every person except identical twins has a unique pattern of fragments.

It is used to identify criminals, settle paternity disputes and identify disaster victims.

For what purpose have bacteria, plants and animals been genetically altered?

Bacteria are altered to produce medicines like human insulin and growth hormone cheaply.

Plants are altered for insect resistance, herbicide tolerance and better nutrition.

Animals are altered to produce medicines in milk and to serve as models of human disease.

Genetic alteration thus improves health, food production and research.

Exercise Q1 — Fill in the blanks (continued: 4-5)

The production of many identical copies of a gene is called gene cloning .

A plasmid is a self-duplicating ring of accessory DNA in the cytoplasm of bacteria.

Exercise Q4 — Extensive questions (continued)

Q4(1). What is the methodology for producing recombinant DNA to be used in gene cloning?

Recombinant DNA is made by isolating the gene of interest and a vector (usually a plasmid) and cutting both with the same restriction enzyme, so they have matching sticky ends.

The gene is joined into the vector with DNA ligase, forming recombinant DNA.

The recombinant plasmid is introduced into a host bacterium by transformation; host cells that took up the plasmid are selected (for example by antibiotic resistance) and grown into clones.

The cloned gene can then be expressed to produce its protein, for example human insulin.

Q4(2). What is a genomic library? How would you locate a gene of interest in the library?

A genomic library is a collection of clones (in bacteria or phages) that together contain all the DNA fragments of an organism's genome.

It is made by cutting the whole genome with restriction enzymes and inserting every fragment into vectors, so each clone carries a different piece.

A gene of interest is located by screening the library with a probe: a labelled single-stranded DNA sequence complementary to the gene, which hybridises only to the clone carrying that gene.

Alternatively, clones can be screened for the protein product of the gene.

Chapter 23 Q2 — flagged, not shown on the exercise pages.

FLAGGED — BOOK NUMBERING NOTE: the Chapter 23 exercise pages as extracted show Q1, Q3 and Q4; no Q2 appears on those pages.

Q2 is therefore not guessed here; this is a render-verified book numbering skip in the book.

Chapter 24: Evolution

Exercise Q1 — Fill in the blanks

Archaebacteria can tolerate high temperatures up to 110 C .

The first eukaryote appeared about 1.5 billion years ago.

Darwin presented the theory of the origin of species by means of natural selection.

Wallace developed a theory of natural selection essentially identical to Darwin's.

Prokaryotes are considered to be the ancestors of all life.

A respiratory protein called cytochrome c is found in all aerobic organisms.

The total aggregate of genes in a population at any time is called its gene pool .

The Hardy-Weinberg theorem describes a non-evolving population.

Evolution is a series of changes in the genetic composition of a population over time.

The level of classification between species and family is called genus .

The Hardy-Weinberg equation is the binomial expansion of (p+q)2 .

An endangered species is in imminent danger of extinction throughout its range.

A deme is a localized group of individuals belonging to the same species.

The first photosynthetic organisms used hydrogen sulphide as a source of hydrogen for reducing carbon dioxide to sugars.

Malthus published an essay on 'The Principle of Population'.

Exercise — Short questions (selection)

State the Hardy-Weinberg theorem and its conditions.

The theorem states that allele frequencies in a population remain constant from generation to generation.

This holds if there is no mutation, no migration, no natural selection and random mating.

The population must also be very large, so chance changes are negligible.

Under these conditions, genotype frequencies follow p2 + 2pq + q2 = 1.

How does natural selection bring about evolution?

Individuals in a population vary, and more are born than can survive.

Those with characters suited to the environment survive and reproduce more.

Their favourable alleles increase in the population over generations.

Gradually the population changes, which is evolution by natural selection.

Give two evidences of evolution from the fossil record.

Fossils show simple organisms in older rocks and more complex forms in younger rocks.

Transitional fossils, like Archaeopteryx with both reptile and bird features, link groups.

The sequence of horse fossils shows gradual change in size and toe number.

Fossils therefore document descent with modification over millions of years.

Exercise Q2 — Short questions (Q2(8) Hardy-Weinberg theorem already present in existing notes)

Q2(1). What are hydrothermal vents?

Hydrothermal vents are cracks in the ocean floor, usually at mid-ocean ridges, from which hot, mineral-rich water escapes.

They support communities of bacteria, tube worms and other animals that live without sunlight, using chemosynthesis (energy from hydrogen sulphide) instead of photosynthesis.

They are important in evolution because early life may have originated in such hot, chemically rich environments.

Q2(2). State the endosymbiont hypothesis.

The endosymbiont hypothesis states that mitochondria and chloroplasts were once free-living prokaryotic cells that were engulfed by a larger host cell and lived inside it in symbiosis.

Evidence is that both organelles have their own circular DNA, their own ribosomes like bacterial ribosomes, double membranes, and divide by binary fission like bacteria.

Over time, most of their genes moved to the nucleus, but they kept enough independence to support the hypothesis.

Q2(3). Define population genetics.

Population genetics is the study of the genetic composition of populations, especially allele frequencies, and how they change over time.

It studies the forces that change allele frequencies: mutation, natural selection, migration (gene flow), genetic drift and non-random mating.

It provides the mathematical basis of evolution, including the Hardy-Weinberg theorem.

Q2(4). How does the fossil record provide evidence of evolution?

The fossil record provides evidence of evolution because fossils appear in a definite order in rock layers: simple organisms in older rocks, more complex forms in younger rocks.

Transitional fossils, such as Archaeopteryx (between reptiles and birds), show intermediate stages.

Extinct forms, and the gradual change of groups such as horses over millions of years, show that life has changed over time.

Q2(5). Explain the term homology with a suitable example.

Homology means similarity of structure due to common ancestry, even when the structures are used differently.

For example, the forelimbs of humans, bats, whales and cats all have the same basic bone plan (humerus, radius, ulna, carpals, digits), although they are used for grasping, flying, swimming and walking.

This shared plan is best explained by inheritance from a common tetrapod ancestor, not by separate creation for each function.

Q2(6). What are vestigial organs? Give two examples.

Vestigial organs are reduced structures that have lost most or all of their original function during evolution, but remain in the body.

Two examples in humans are the appendix (a small remnant of the caecum) and the coccyx (tail bones).

Whales retain small pelvic bones, and flightless birds retain reduced wings, which are also vestigial.

Q2(7). How are evolutionary relationships reflected in DNA and proteins?

Evolutionary relationships are reflected in DNA and protein sequences: the more similar two species' sequences are, the more recently they shared a common ancestor.

For example, the protein cytochrome c is found in all aerobic organisms, and its amino-acid sequence differs by only a few residues between closely related species but by many between distant ones.

DNA hybridisation and genome comparisons give the same pattern, confirming relationships suggested by anatomy and fossils.

Q2(9). What is the difference between endangered species and threatened species?

An endangered species is in imminent danger of extinction throughout its range, for example the Indus dolphin and the common leopard in Pakistan.

A threatened species is likely to become endangered in the near future if the causes of its decline continue, but is not yet in imminent danger.

Endangered is therefore the more urgent category, and threatened is the warning stage before it.

Q2(10). Name any five species declared extinct in Pakistan.

Five species declared extinct in Pakistan, as listed in the chapter, are: cheetah, tiger, Asian lion, Indian rhinoceros and cheer pheasant.

The chapter also lists crocodile and gavial (gharial) as extinct in Pakistan.

Their extinction was caused mainly by habitat destruction, hunting and human disturbance.

Exercise Q4 — Extensive questions

Q4(1). What are the endangered species? What measures could be adapted for their

preservation?

An endangered species is one in imminent danger of extinction throughout its range.

Preservation needs a multifaceted plan: a global system of national parks and wildlife corridors to protect large habitats and allow movement.

Protected landscapes and multiple-use areas allow controlled human activity while keeping wildlife value; zoos and botanical gardens save species whose extinction is imminent through captive breeding.

Laws against hunting and trade, public awareness, and protection of critical habitats such as wetlands and deserts are also essential.

Q4(3). Describe evidences of evolution from any five branches of biology.

Five branches of biology give evidence of evolution. (1) Fossil record: ordered succession of forms and transitional fossils such as Archaeopteryx.

(2) Comparative anatomy: homologous structures (forelimbs of vertebrates) show common ancestry, and analogous structures show adaptation.

(3) Embryology: vertebrate embryos share gill slits and tails early in development, showing common descent.

(4) Biogeography: related species are found in nearby regions, and isolated islands have unique species, as Darwin saw in the Galapagos.

(5) Molecular biology: DNA and protein sequences (for example cytochrome c) are more similar in closely related species, confirming evolutionary trees.

Q4(4). How did evolution proceed from prokaryotes to eukaryotes? Analyse Darwin's theory of natural selection as a mechanism of evolution.

Prokaryotes were the first life; eukaryotes evolved from them when a host cell engulfed aerobic bacteria (which became mitochondria) and, later, photosynthetic bacteria (which became chloroplasts).

Infolding of the cell membrane may have formed the nucleus and endomembrane

system, giving a complex cell with organelles.

Darwin's theory of natural selection as a mechanism is: populations produce more offspring than can survive; individuals vary; those with variations suited to the environment survive and reproduce (survival of the fittest); and the favourable variations are inherited, so the population changes over generations.

Chapter 24 Q3 — flagged, not shown on the exercise pages.

NOTE: extensive item (2) (Hardy-Weinberg theorem) is already covered in the existing notes and is not duplicated here.

FLAGGED — render-verified book numbering skip: the Chapter 24 exercise pages as extracted show Q1, Q2 and Q4; no Q3 appears on those pages.

Chapter 25: Ecosystem

Exercise Q1 — Fill in the blanks

A group of similar organisms living together in space and time is called a population .

Organisms which can synthesize their own food are called producers .

Animals, non-green plants and microorganisms directly or indirectly depend upon green plants for their food, so they are called consumers .

Exercise Q2 — True or False

At different places in an environment, when you study only one population, it will be synecology.

False

Abiotic components include all living components.

False

Primary succession starting in a pond is called xerosere.

False

The animal that is caught and eaten is the predator.

False

Endoparasites live inside the body of the host.

True

Exercise Q4 — Short questions

What are biogeochemical cycles?

Biogeochemical cycles are the circulation of elements between living things and the environment.

Carbon, nitrogen, water and phosphorus move through air, water, soil and organisms.

Decomposers return elements from dead bodies to the soil and air.

These cycles keep essential nutrients available for new generations of life.

Sketch three main steps in the nitrogen cycle.

First, nitrogen-fixing bacteria convert atmospheric nitrogen into ammonia and nitrates.

Second, plants absorb nitrates and animals obtain nitrogen by eating plants.

Third, decomposers return nitrogen from dead matter to the soil as ammonia.

Denitrifying bacteria finally return nitrogen gas to the atmosphere, completing the cycle.

Define grazing.

Grazing is feeding by herbivores on plants, especially grasses.

The animal eats parts of plants without necessarily killing them.

Grazing transfers energy from producers to primary consumers.

Overgrazing can destroy vegetation and cause soil erosion.

What percentage of sun energy reaches plants?

Only about one percent of the sun's energy reaching the earth is captured by plants.

Most energy is reflected, absorbed by clouds or lost as heat.

Plants use the captured energy for photosynthesis to make food.

About ten percent of this stored energy then passes to each higher trophic level.

What is autecology? What is synecology?

Autecology is the study of a single species in relation to its environment.

It examines how one population adapts to temperature, water and food.

Synecology is the study of whole communities of different species together.

It examines how populations interact in food webs and share a habitat.

Exercise Q5 — Extensive questions

Q5(1). Define environment. What must environment supply for insects, green plants, birds, animals and people?

The environment is the sum of all physical (abiotic) and biological (biotic) factors surrounding an organism.

It must supply food (energy), water, oxygen (for animals) or carbon dioxide (for plants), shelter and space, and a suitable temperature and light range.

Insects need food plants, water and shelter; green plants need light, water, carbon dioxide and minerals; birds and animals need food, water, air and nesting or living space; people need all of these, plus a healthy, unpolluted environment.

Q5(2). What factors in the environment can affect all living things? Are they important to survive in a biome?

Environmental factors that affect all living things are light, temperature, water, air, soil and the other organisms around them.

They are important for survival in a biome because each organism can live only

within a range of each factor; beyond that range it cannot survive or reproduce.

For example, temperature and rainfall together decide whether a region is desert, grassland or forest, and which species can live there.

Q5(3). What can you conclude about all the physical and biological factors in an environment?

Physical and biological factors in an environment are interdependent: climate, soil and water shape the community of plants and animals, and the organisms in turn change the soil, air and water.

No factor acts alone; a change in one, such as rainfall, affects food, shelter and population size throughout the community.

An ecosystem is therefore a web of interactions, not a collection of separate parts.

Q5(4). What is biosphere? What must the biosphere provide for living things? Why is a biosphere absent on the moon?

The biosphere is the part of the earth where life exists: the lower atmosphere, the oceans and surface waters, and the upper soil and rocks.

It must provide liquid water, breathable air, food (energy from the sun through producers), suitable temperatures, and protection from harmful radiation.

A biosphere is absent on the moon because the moon has no atmosphere, no liquid water, extreme temperatures (very hot by day, very cold by night), and no protection from solar and cosmic radiation.

Q5(5). Define succession. Discuss succession on land.

Succession is the gradual, orderly change in the species of a community over time, until a stable climax community is reached.

Succession on land (xerosere) begins on bare rock: lichens and mosses break down rock and build soil, then herbs and grasses grow, followed by shrubs, and finally trees form a climax forest.

Each stage changes the soil and microclimate, making it suitable for the next stage

and unsuitable for itself, until the climax community is stable.

Chapter 25 Q3 — flagged, not shown on the exercise pages.

FLAGGED — BOOK NUMBERING NOTE: the Chapter 25 exercise pages as extracted show Q1, Q2, Q4 and Q5; no Q3 appears on those pages.

Q3 is therefore not guessed here; this is a render-verified book numbering skip in the book.

Chapter 26: Some Major Ecosystems

Exercise Q1 — Fill in the blanks

Water is slower to heat and cool than air.

The distribution of life in lakes depends on access to light and to places for attachment.

An ecosystem on land is also known as a terrestrial ecosystem.

An ecosystem in water is also called an aquatic ecosystem.

Exercise Q2/Q4 — Short and extensive questions

Define productivity of an ecosystem.

Productivity is the rate at which an ecosystem produces organic matter.

Gross primary productivity is the total food made by producers through photosynthesis.

Net primary productivity is what remains after producers use some food for respiration.

It is measured as grams of dry matter per square metre per year.

List four adaptations in plants and animals for a terrestrial ecosystem.

Terrestrial plants have roots to absorb water and a waxy cuticle to prevent water

loss.

They have stomata that close in dry conditions and strong stems to stand upright.

Terrestrial animals have lungs for breathing air and waterproof skin or scales.

They also have limbs for movement on land and kidneys that conserve water.

Name three zones in a lake ecosystem.

The littoral zone is the shallow shore region where light reaches the bottom and plants grow.

The limnetic zone is the open sunlit surface water where plankton live.

The profundal zone is the deep, dark, cold region where no photosynthesis occurs.

Decomposers and bottom feeders dominate the profundal zone.

How many biomes are present in the world? Name five of them.

There are seven major terrestrial biomes in the world.

Five of them are: tundra, taiga (coniferous forest), temperate deciduous forest.

Grassland, desert and tropical rain forest complete the list with chaparral.

Each biome has its own climate, soil and characteristic plants and animals.

Give the names of some major ecosystems on land in Pakistan.

Pakistan has the Indus delta and mangrove forests along the coast.

It has the Thar and Cholistan deserts with drought-adapted plants and animals.

The northern alpine meadows and coniferous forests are rich in wildlife.

The Punjab plains and the Potohar plateau form the main agricultural ecosystems.

What are the four major requirements for life? Which two are limiting in a terrestrial ecosystem?

The four requirements for life are water, food (energy), oxygen and a suitable temperature.

In terrestrial ecosystems, water is often the most limiting factor.

Suitable temperature is the second limiting factor, especially in deserts and cold regions.

Food and oxygen are usually available, but water and temperature decide where life can thrive.

Exercise Q4 — Extensive questions

Q4(2). List some adaptations of (a) desert plants and (b) desert animals to heat and drought.

(a) Desert plants: deep or widespread roots to reach water, thick waxy cuticles and reduced leaves or spines to cut water loss, water-storing stems (cacti), and CAM photosynthesis, opening stomata at night.

(b) Desert animals: they are mostly nocturnal and shelter in burrows during the heat; they conserve water with concentrated urine and dry faeces, and obtain water from food and metabolism.

Some, like the kangaroo rat, never drink; others store fat (camel hump) and tolerate large changes in body temperature.

Q4(3). Where is life in oceans (hydrospheric ecosystem) most abundant and why?

Life in the oceans is most abundant in the shallow coastal waters, especially over the continental shelf and in the photic (sunlit) zone.

Light reaches these waters, so photosynthesis by phytoplankton supports the whole food web, and nutrients washed from the land are plentiful.

Warm temperatures and varied habitats (reefs, estuaries, kelp beds) also support more species than the deep, dark open ocean.

Q4(4). Distinguish between three different zones in the lake ecosystem.

The littoral zone is the shallow shore region where light reaches the bottom, rooted plants grow, and many animals live among the plants.

The limnetic zone is the open, sunlit surface water away from the shore, where plankton float and most photosynthesis occurs, but rooted plants are absent.

The profundal zone is the deep, dark bottom water below the light, where no

photosynthesis occurs; it is cold, low in oxygen, and inhabited mainly by decomposers and bottom feeders.

Chapter 26 Q3 — flagged, not shown on the exercise pages.

NOTE: extensive item (1) (four major requirements for life) is already covered in the existing notes and is not duplicated here.

FLAGGED — render-verified book numbering skip: the Chapter 26 exercise pages as extracted show Q1, Q2 and Q4; no Q3 appears on those pages.